



**Year 11 Mathematics Advanced
Preliminary HSC Assessment Task 1
Term 2, Week 3, 2019 VUL**

Name: _____

Wednesday 15th May 2019

- Answer all questions on the lined paper provided.
- Start each question on a new page.
- NESA approved calculators are permitted in this task.
- Marks may be deducted for insufficient or illegible work.
- Total possible mark: **50**
- Time allowed: **80 minutes + 5 minutes reading time**

Question 1 (8 marks)	Marks
(a) Solve $(x + 4)^2 = 9$.	1
(b) Simplify $\frac{2\sqrt{6} \times 3\sqrt{7}}{12\sqrt{21}}$.	1
(c) Factorise each expression completely.	
(i) $9x^2 - 6x - 8$.	1
(ii) $l^2 - m^2 + 2mn - n^2$.	2
(d) Find the exact solutions of $\frac{5m}{2} = 2 + \frac{1}{m}$.	2
(e) A cyclist rides for 5 hours at a certain speed and then for 4 hours at a speed 6 km/h greater than his original speed. If he rides 294 altogether, what was his original speed?	1

End of Question 1

Question 2 starts on the next page

Question 2 (8 marks) *Start a new page***Marks**

(a) State the domain of the following functions:

(i) $y = \sqrt{1-3x}$ **1**

(ii) $y = \frac{1}{x^3+1}$. **1**

(b) By the use of a sketch, or otherwise, state the range of $y = -\sqrt{4-x^2}$. **1**

(c) State the definition of a rational number. **1**

(d) Without using a calculator, show that the following expression is rational: **2**

$$\frac{1}{3+\sqrt{6}} + \frac{2}{\sqrt{6}} .$$

(e) Solve, by the substitution method, these simultaneous equations: **2**

$$y = 3x^2$$

$$y = x .$$

End of Question 2

Question 3 starts on the next page

Question 3 (8 marks) *Start a new page*

Marks

- (a) If $f(x) = \sqrt{1-x^2}$, explain why $f(-2)$ cannot be found. **1**
- (b) If $P(x) = x^2 - 2x - 4$, find $P(\sqrt{3}-1)$ in simplest exact form. **1**
- (c) Solve $|1-3x|=2$ for x . **2**
- (d) Sketch the parabola $y = -x^2 + 8x - 15$ clearly indicating its vertex and all intercepts with the coordinate axes. **2**
- (e) If $g(x) = x^2 + 5x$, find $\frac{g(a)-g(c)}{a-c}$ in simplest form. **2**

End of Question 3

Question 4 starts on the next page

Question 4 (8 marks) *Start a new page*

Marks

- (a) Sketch the graph of $xy = 4$ and classify it as one-to-one, many-to-one, one-to-many or many-to-many. Justify your classification. 2
- (b) Consider the polynomial $y = (1 - x)(x + 1)^2(x - 2)^3$.
- (i) Write down the zeroes of the polynomial. 1
- (ii) By using a table of values to test its sign, or otherwise, sketch the polynomial. 2
- (c) The wavelength λ in metres of a musical tone is inversely proportional to its frequency f in vibrations per second.
- (i) Express this relationship algebraically. 1
- (ii) The frequency of the musical note “middle C” is 260 vibrations per second and its wavelength is 1.319 metres. Find the wavelength of a musical note with frequency 440 vibrations per second, correct to 3 decimal places. 2

End of Question 4

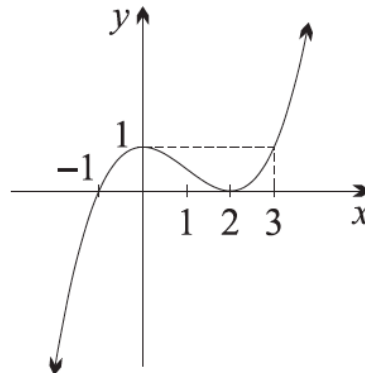
Question 5 starts on the next page

Question 5 (8 marks) *Start a new page***Marks**

- (a) If $f(x) = x^3 - x$, find $f(-x)$ and hence determine whether the function is odd, even or neither.

1

- (b) The graph shown below is the curve $y = f(x)$.



Use this graph to sketch separate graphs of:

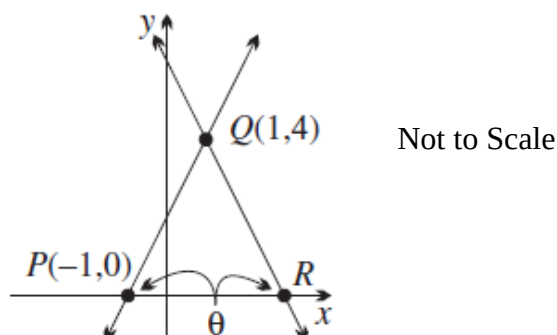
- (i) $y = f(x-1)$ **1**
- (ii) $y = -f(-x)$. **1**
- (c) Sketch $x^2 - 10x + y^2 + 8y + 32 = 0$. **2**
- (d) (i) Sketch $y = |x+1| - 1$ carefully showing the intercepts with the coordinate axes. **2**
- (ii) For what values of k does $|x+1| = k+1$ have no solutions for x ? **1**

End of Question 5

Question 6 starts on the next page

Question 6 (10 marks) *Start a new page***Marks**

- (a) The vertices of the triangle in the diagram below are $P(-1,0)$, $Q(1,4)$ and R , where R lies on the x -axis and $\angle QPR = \angle QRP = \theta$.



- | | | |
|-------|--|---|
| (i) | Find the midpoint of PQ . | 1 |
| (ii) | Find the size of θ correct to the nearest degree. | 1 |
| (iii) | Find the equation of the line PQ . | 1 |
| (iv) | Explain why the gradient of line QR is -2 . | 1 |
| (v) | By considering the area of $\triangle PQR$, find the shortest distance from P to QR . | 2 |
- (b) If $L(x) = x + 1$ and $Q(x) = x^2 + 2x$;
- | | | |
|-------|---|---|
| (i) | Find the zeros of $Q\left(L\left(\frac{1}{x+1}\right)\right)$. | 1 |
| (ii) | Find the equation of the horizontal asymptote of $y = Q\left(L\left(\frac{1}{x+1}\right)\right)$. | 1 |
| (iii) | It is known that as $x \rightarrow -1^-$ and as $x \rightarrow -1^+$, $y \rightarrow \infty$. (Do not prove this)
By using a table of values to test the sign of the function,
or otherwise, sketch the graph of $y = Q\left(L\left(\frac{1}{x+1}\right)\right)$, $x \neq -1$. | 2 |

End of Assessment

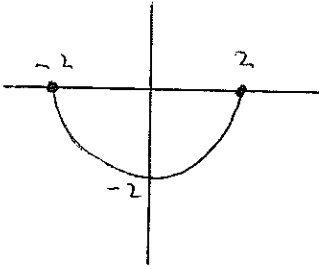


2019 Year 11 Mathematics Advanced Task 1 SOLUTIONS

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 1 (a) $(x+4)^2 = 4$ $x+4 = \pm 2$ $x = -1, -7 \quad \checkmark$ - correct answers (b) $\frac{2\sqrt{6} \times 3\sqrt{7}}{12\sqrt{21}}$ $= \frac{6\sqrt{6}}{12\sqrt{3}}$ $= \frac{\sqrt{2}}{2} \quad \checkmark$ - must be fully simplified (c) (i) $9x^2 - 6x - 8$ $= (3x-4)(3x+2) \quad \checkmark$ (ii) $d^2 - (m^2 - 2mn + n^2)$ $= d^2 - (m-n)^2 \quad \checkmark$ $= (d-m+n)(d+m-n) \quad \checkmark$ [AWO for $(d-m)(d+m) + n(2m-n)$]		(d) $\frac{5m}{2} = 2 + \frac{1}{m}$ $\frac{5m^2}{2} = 2m + 1$ $5m^2 - 4m - 2 = 0 \quad \checkmark$ $m = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(5)(-2)}}{2(5)} \quad \checkmark$ $= \frac{4 \pm \sqrt{56}}{10}$ $= \frac{4 \pm 2\sqrt{14}}{10}$ $= \frac{2 \pm \sqrt{14}}{5}$ (e) $s = \frac{d}{t}$ $\therefore d = s \times t$ Let initial speed be s $5s + 4(s+6) = 294$ $9s + 24 = 294$ $9s = 270$ $\therefore s = 30 \text{ km/h} \quad \checkmark$	

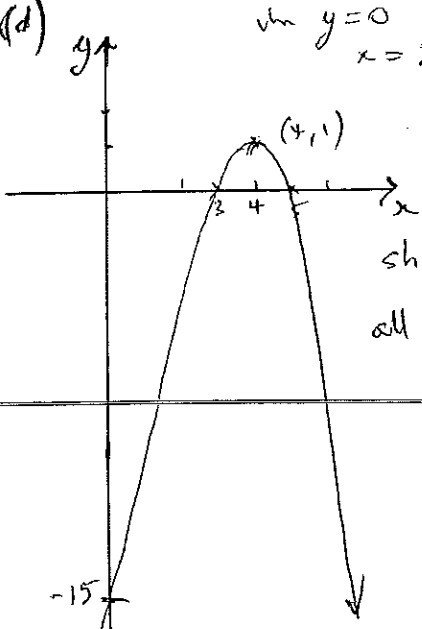


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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 2 (a) (i) $1 - 3x \geq 0$ $-3x \geq -1$ D: $x \leq \frac{1}{3}$ ✓ (ii) $x^3 + 1 \neq 0$ $x \neq -1$ ✓ ∴ D: All real $x \neq -1$ (b)  ∴ R: $-2 \leq y \leq 0$ ✓ (c) A rational number can be expressed in the form $\frac{p}{q}$ where $p, q \neq 0$ are integers with no common factor other than 1. ✓		(d) $\frac{1}{3 + \sqrt{6}} + \frac{2}{\sqrt{6}}$ $= \frac{3 - \sqrt{6}}{9 - 6} + \frac{2\sqrt{6}}{6}$ ✓ $= \frac{6 - 2\sqrt{6} + 2\sqrt{6}}{6}$ $= 1$ ✓ which is rational. (e) $y = 3x^2$ $y = x$ $3x^2 - x = 0$ $x(3x - 1) = 0$ $\therefore x = 0 \text{ or } \frac{1}{3}$ ✓ $y = 0 \text{ or } \frac{1}{3}$ ✓	

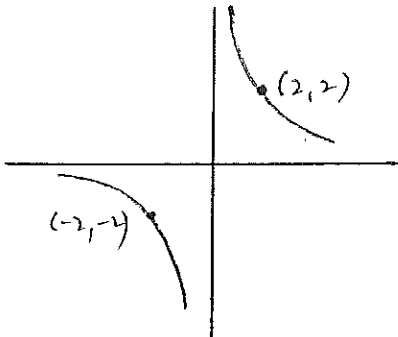
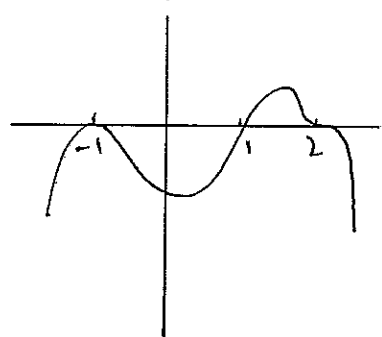


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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 3 (a) $f(-2) = \sqrt{1-(-2)^2}$ $= \sqrt{-3}$ which is not real. (You can't take the square root of a negative number). (b) $p(\sqrt{3}-1) = (\sqrt{3}-1)^2 - 2(\sqrt{3}-1) - 4$ $= 4 - 2\sqrt{3} - 2\sqrt{3} + 2 - 4$ $= 2 - 4\sqrt{3} \quad \checkmark$ (c) $1-3x = 2$ $1-3x = \pm 2 \quad \checkmark$ $3x = -1 \quad 3x = 3$ $x = -\frac{1}{3} \quad x = 1 \quad \checkmark$ both answers (d) $y = x^2 - 8x + 15$ $x = 3 \text{ or } 5$  shape and vertex $\checkmark$ all intercepts $\checkmark$		(a) $g(x) = x^2 + 5x$ $\frac{g(a)-g(c)}{a-c} = \frac{a^2+5a-(c^2+5c)}{a-c} \quad \checkmark$ $= \frac{a^2-c^2+5a-5c}{a-c}$ $= \frac{(a-c)(a+c)+5(a-c)}{a-c}$ $= \frac{(a-c)(a+c+5)}{a-c}$ $= a+c+5 \quad \checkmark$	

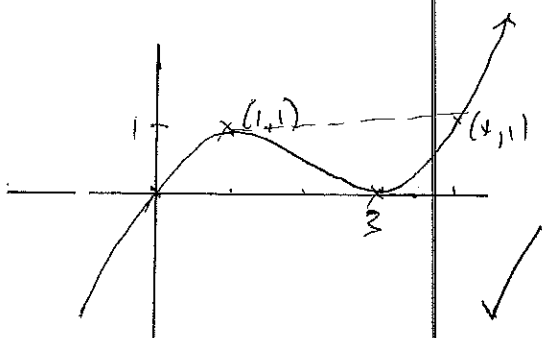
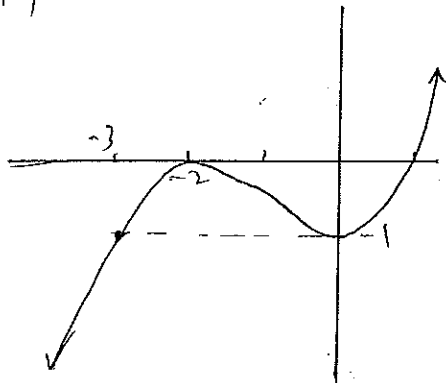
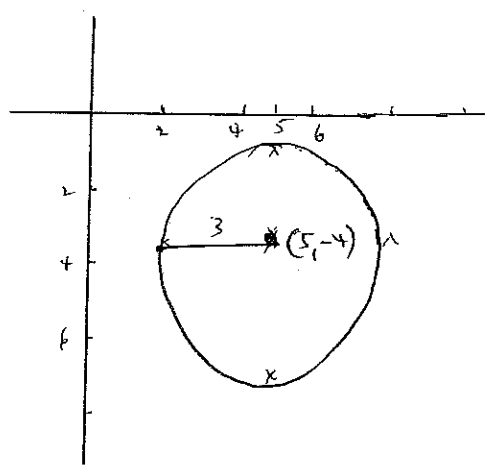
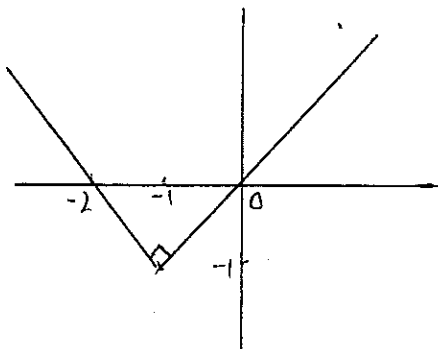


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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 4 (a)  This is a "one-to-one" function as there is 1 x-value for each y-value in its range (b)  ✓ - intercepts at zero's and $y \rightarrow -\infty$ for extreme values at x ✓ - correct shape at zeros.	✓ graph ✓ correct classification with reason.	c) i) $\lambda \propto \frac{1}{f}$ ✓ either or $\lambda = \frac{k}{f}$ ii) $1.319 = \frac{k}{260}$ $\therefore k = 260 \times 1.319$ ✓ $\therefore \lambda = 440$ $\lambda = \frac{260 \times 1.319}{440}$ $= 0.779 \text{ m.}$ ✓	



2019 Year 11 Mathematics Advanced Task 1 SOLUTIONS

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Question 5 (a) $f(x) = x^3 - x$ $f(-x) = (-x)^3 - (-x)$ $= -x^3 + x$ $= -(x^3 - x)$ ✓ with correct conclusion. $= -f(x)$ $\therefore f(x)$ is ODD (b) (i)  (ii) 		c) $x^2 - 10x + 25 + y^2 + 8y + 16$ $= -32 + 25 + 16$ $\therefore (x-5)^2 + (y+4)^2 = 9$ $\therefore C(5, -4) r = 3$  d) (i)  ✓ the "V" shape with either one of the translations ✓ correct solution. (ii) $x+1 - 1 = k$ no solutions if $k < -1$ ✓	



2019 Year 11 Mathematics Advanced Task 1 SOLUTIONS

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Question 6 (a) (i) $M\left(\frac{-1+1}{2}, \frac{0+4}{2}\right)$ $= M(0, 2)$ (ii) $\tan \theta = m_{PQ}$ $= \frac{4-0}{1-(-1)}$ $\tan \theta = 2$ $\therefore \theta \doteq 63^\circ$ nearest degree (iii) $y - 4 = 2(x - 1) \checkmark$ $y = 2x + 2$ or equivalent (iii) Since $\angle QRP = \theta = 63.43\dots^\circ$ $m_{QR} = \tan(180^\circ - 63.43\dots^\circ)$ $= -2$ or equivalent algebraic argument (iv) Since $\triangle PQR$ is isosceles $\text{Area} = \left(\frac{1}{2} \times 2 \times 4\right) \times 2$ $= 8 \text{ u}^2 \checkmark$		Equation of QR $y - 4 = -2(x - 1)$ $y = -2x + 6$ thus R is $(3, 0)$ $d_{QR} = \sqrt{(3-1)^2 + (0-4)^2}$ $= \sqrt{4 + 16}$ $= \sqrt{20} = 2\sqrt{5}$ Shortest distance (d) is perpendicular distance thus $\frac{1}{2} \times QR \times d = 8$ $\frac{1}{2} \times 2\sqrt{5} \times d = 8$ $\therefore d = \frac{8}{\sqrt{5}} \checkmark$ units.	



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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments																				
Question 6 (b) $L(x) = x + 1$ $Q(x) = x^2 + 2x$ (i) $L\left(\frac{1}{x+1}\right) = \frac{1}{x+1} + 1$ $= \frac{x+2}{x+1}$ $\therefore Q\left(\frac{1}{x+1}\right) = Q\left(\frac{x+2}{x+1}\right)$ $= \left(\frac{x+2}{x+1}\right)^2 + 2\left(\frac{x+2}{x+1}\right)$ $= \frac{(x+2)^2}{(x+1)^2} + \frac{2(x+2)(x+1)}{(x+1)^2}$ $= \frac{(x+2)[(x+2) + 2(x+1)]}{(x+1)^2}$ $= \frac{(x+2)(3x+4)}{(x+1)^2}$ which has zeros $x = -2, -\frac{4}{3}$ ✓ (ii) as $x \rightarrow \infty, y \rightarrow 3^+$ $x \rightarrow -\infty, y \rightarrow 3^-$ $\therefore$ Horizontal asymptote is $x = 3$ ✓		iii) <table border="1"> <tr> <td>x</td> <td>$-\infty$</td> <td>-3</td> <td>-2</td> <td>-1.5</td> <td>$-1\frac{1}{3}$</td> <td>-1</td> <td>-1</td> <td>-1^+</td> <td>∞</td> </tr> <tr> <td>y</td> <td>3^-</td> <td>+</td> <td>0</td> <td>-</td> <td>0</td> <td>∞</td> <td>AA</td> <td>∞</td> <td>3^+</td> </tr> </table> ✓ correct behaviour near <u>both</u> $x = -1$ and $y = 3$ ✓ correct <u>sign</u> and shape between zeros and <u>asymptotes</u>.	x	$-\infty$	-3	-2	-1.5	$-1\frac{1}{3}$	-1	-1	-1^+	∞	y	3^-	+	0	-	0	∞	AA	∞	3^+	
x	$-\infty$	-3	-2	-1.5	$-1\frac{1}{3}$	-1	-1	-1^+	∞														
y	3^-	+	0	-	0	∞	AA	∞	3^+														